

MFU Online Switch Monitoring

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ABSTRACT

The MFU online switch monitoring is service for all of internet user in Mae Fah Luang university to monitor the status of devices and location of devices by using the online website. The purpose of this project is to give the advantages not only for user but also staff of Mae Fah Luang as well. Staff of MFU network team can easily monitor the status of devices which are all switch in Mae Fah Luang university are there active or not. To check the status of the switch in this project we are using the Simple Network Management Protocol (SNMP) for communicating between web service and devices.

I. INTRODUCTION

Nowadays the internet is the most important system that allows people communicating across places. No matter how far the distance between the places, the internet will make people to connect to each other and make communication through the internet connection. To allow people to use the internet, it requires many different types of network devices. Also, Mae Fah Luang University contains hundreds of network devices that make student teacher staff and also outsider to access the internet.

Network devices are the backbone of the internet in Mae Fah Luang University. There are many types of network devices of Mae Fah Luang University, which are router, switch layer 2, switch layer 3 or multi-switch, access point and etc. Thus, in this project we are focused on switch layer 3 or the multi-switch devices. This project will be detecting, checking the status and marking the location of all two kinds of switches to make staff or developers easily find the problem and fix the devices in a short time.

II. RELATED WORK

A. MFU WIFI Map

Monitoring the status of all of access point in Mae Fah Luang University by using SNMP. Because when network device down or not online the status of the devices will show only on the management tools that are located only onsite of developer and engineer. For preventing the problem that will occur when developer is out of service or unable to locate at onsite, due the pandemics or the personal issue.

This program will report the status of access point devices in Mae Fah Luang university to help the developer or engineer acknowledge the status of devices. It will help the engineer and developer learn causes of problem and find to easy way to solve it at anywhere that internet can reached. And this program also allow user to check the status of devices.

The status of device will show as real-time online website. The website (Fig1) will show the location of all access point of Mae Fah Luang University and represent the status of access point in 2 different color; Green: Up, Red: Down.

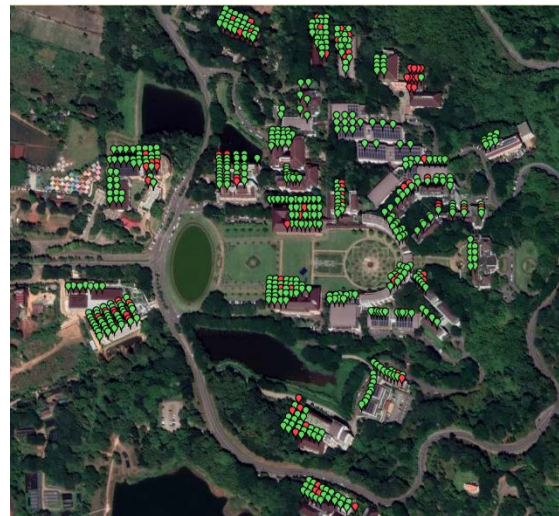


Figure 1 Website of MFU Wi-Fi map

A. Network Switch

The functions of a switch and a router are combined in a switch and a multi-switch. It functions as a switch to link devices on the same subnet or virtual LAN at lightning speeds, and it also includes IP routing intelligence to work as a router. It can support routing protocols, analyze incoming packets, and even route based on source and destination addresses. A layer 3 switch can function as both a switch and a router in this manner.

Feature of switch layer 3.

- Act as switch to connect devices within the same subnet.
- Switching algorithm is simple and is the same for most routed protocols.
- Performs on two OSI layers which are layer 2 and layer 3.



Figure 2

B. SNMP

The Simple Network Management Protocol (SNMP) is a protocol that allows different devices on a network to exchange data. It enables devices to communicate despite the fact that they have distinct hardware and software. Network management tools would be unable to identify devices, monitor network performance, follow network changes, or assess the condition of network devices in real time without the use of a protocol like SNMP. And using SNMPWALK tool to automate request data from SNMP server.

The term snmpwalk is given to an SNMP application that automatically executes several requests. Which request is used to query and obtain SNMP data from a device that is enabled? The snmpwalk command is utilized because it allows users to chain requests together without having to type in separate instructions for each OID or node within a sub-tree.

C. Google Map Javascript API

What is Google Map API?

Customize maps with your own material and graphics for display on web sites and mobile devices using the Maps JavaScript API. The Maps JavaScript API has four basic map types (roadmap, satellite, hybrid, and terrain), each of which can be customized with layers and styles, controls and events, and a variety of services and libraries.

The Google Maps API is a powerful tool that can be used to make a custom map, a searchable map, check-in functions, display live data synching with location, plan routes, and make a mashup.

III. SYSTEM MODEL

The back-end of this project's system is made up of Network Switch and Python programming, while the front-end is made up of website work. To collect all network device information, we need a database, SQL server, to store all of the essential data. To display on the website, the front-end will request information about the devices from a database. And every time the state of devices changes, all of the data in the database will be updated.

A. Retrieving raw data

We utilize snmpwalk, a linux OS tool that sends SNMP requests to devices, to retrieve and collect raw data from switch devices. To apply all of the requests to a large number of network devices, we use python programming to query commands to all available devices. After receiving an SNMP request, network devices will respond with data based on the OID category to which we sent the request (OID is the set of number use as command to communicate through SMNP protocol). Because the data sent by devices via the SNMP protocol may be mixed up, we should filter and clean it before using it.

B. Cleansing and Transform the data

The data we get from a network device is raw data, which is still unclear. We need to clean the raw data to make it usable and beneficial. In this project, we filtered data using Python programming, denying data that we couldn't use and collecting only usable data.

For example, if the device receives a request for traffic utilization, it will provide packages for each

Figure6 Adding new line-connection.

V SUGGESTION

This project can apply to many types of technology for example apply with machine learning for preventing looping of switch. We believe that this project will improve and making solution for MFU staff to monitor network device and MFU student to upgrade and improve this project to be better and make it more useful in many context and situation. However, this project is the beginning of MFU student to develop website about network device the solution that we offer might not compose all the solution for MFU, but we hope this project will be the guideline for future work.

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